

KAN : d=25



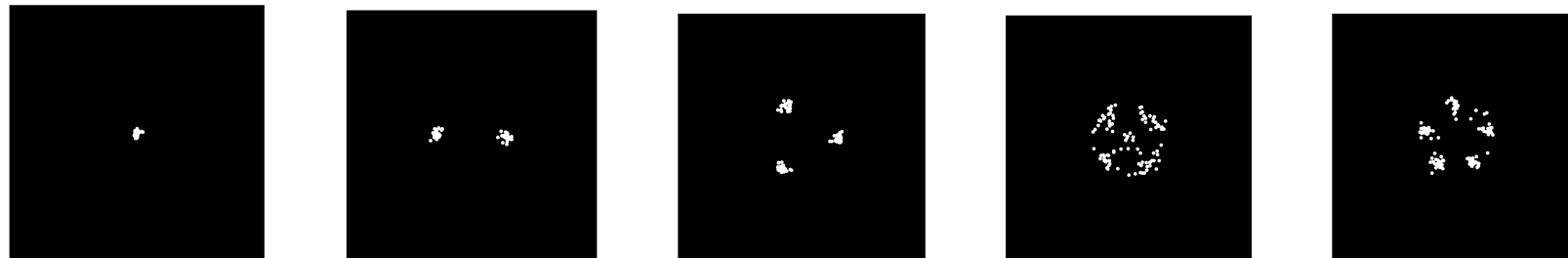
KAN: d=50



KAN: $d=75$



KAN: $d=100$



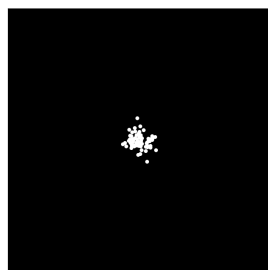
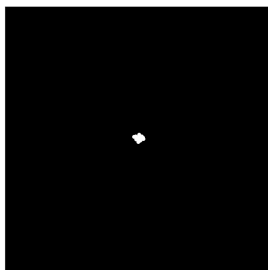
KAN: $d=125$



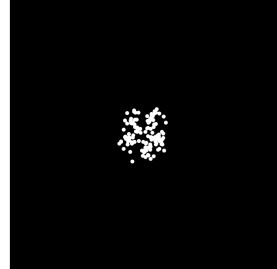
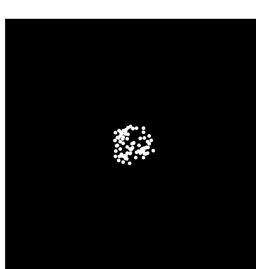
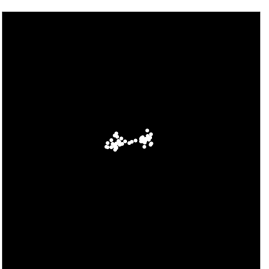
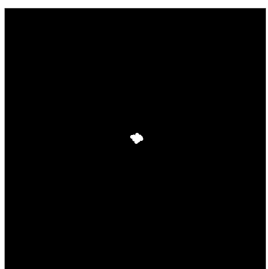
KAN: $d=150$



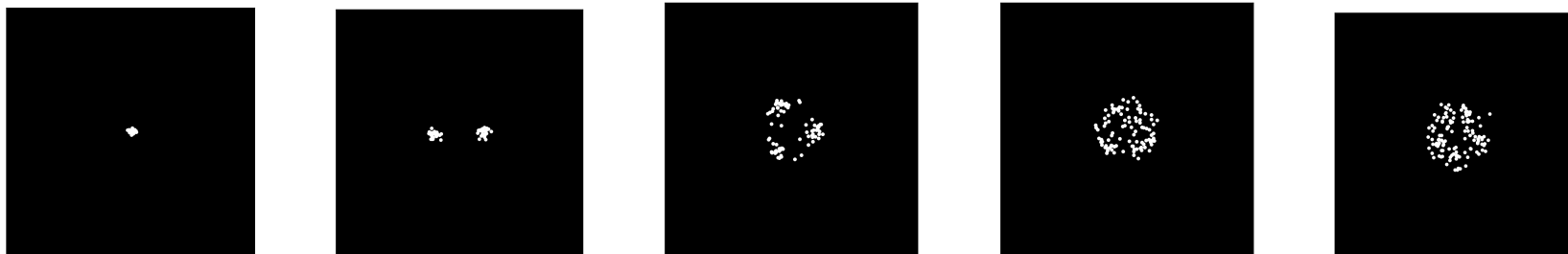
CNN+KAN : d=25



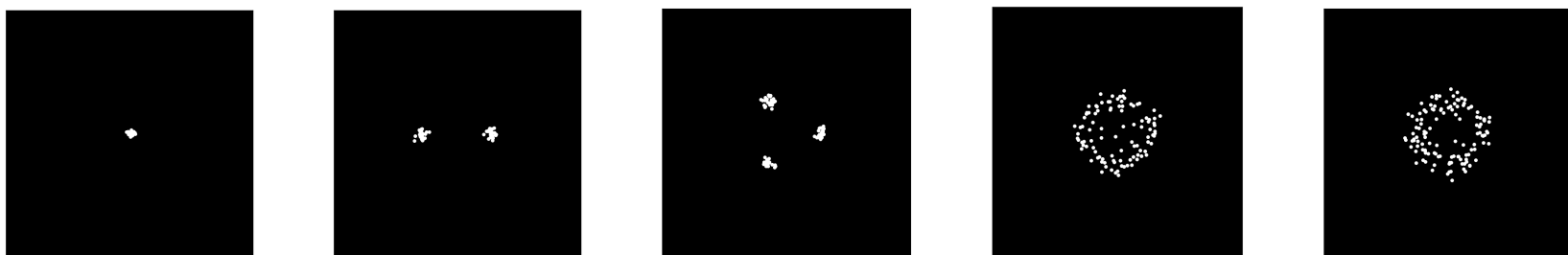
CNN+KAN: d=50



CNN+KAN : d=75



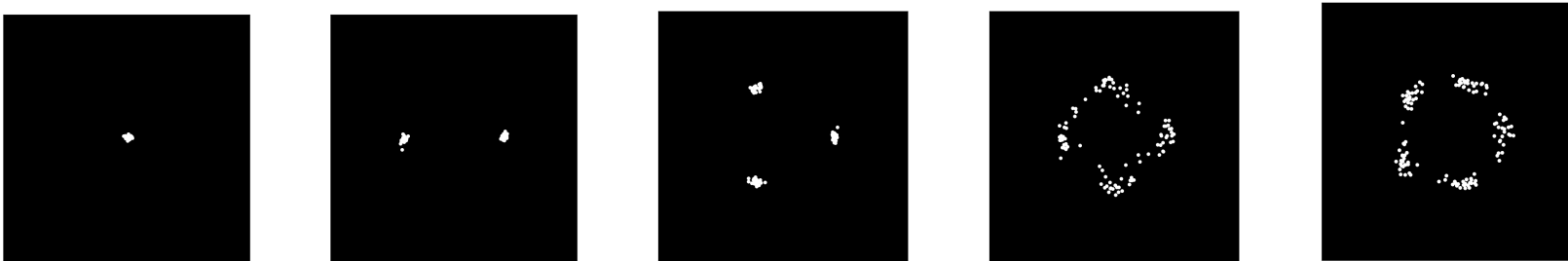
CNN+KAN : d=100



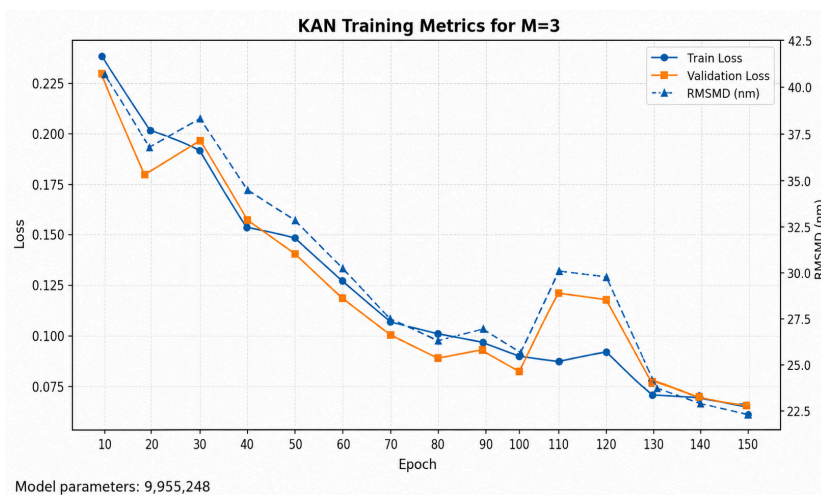
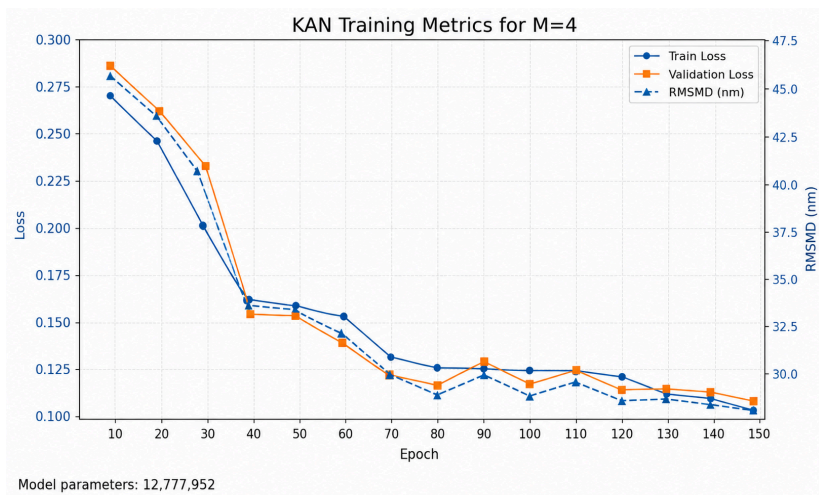
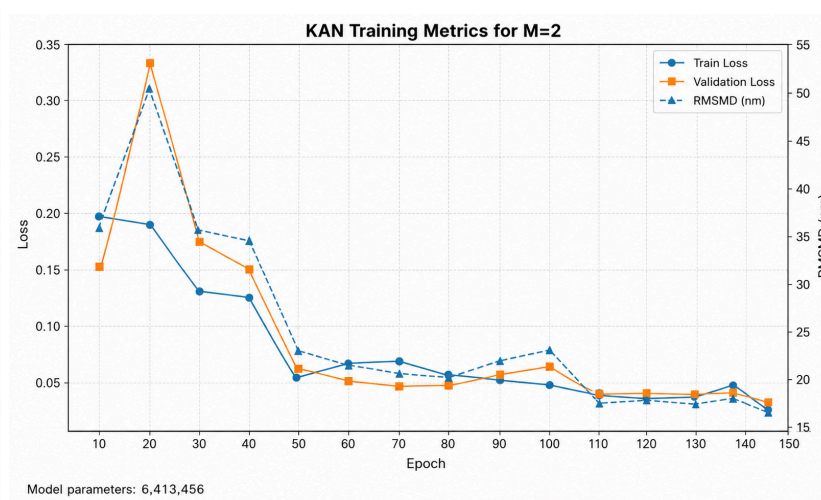
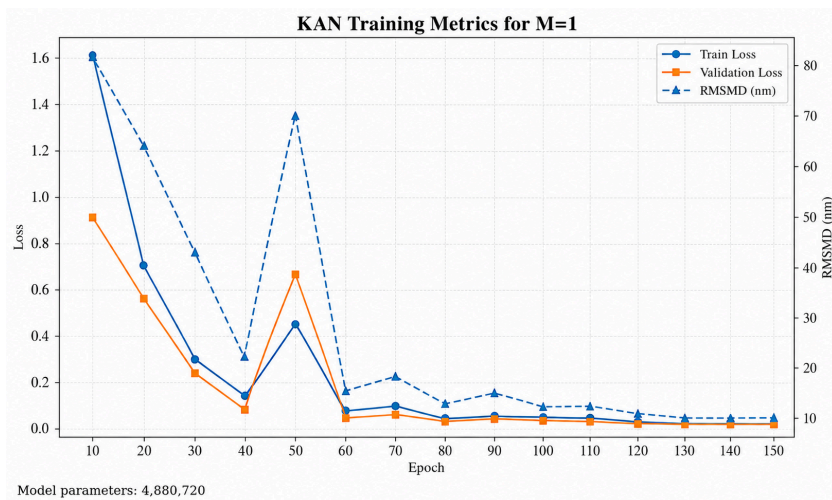
CNN+KAN : d=125

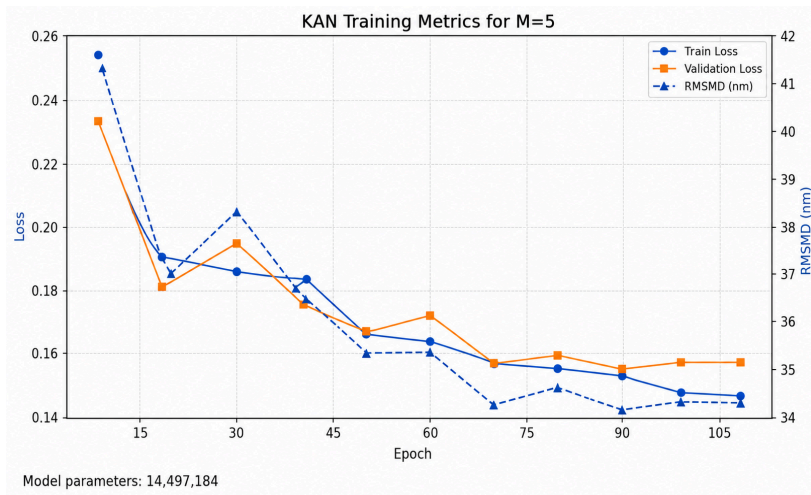


CNN+KAN : d=150

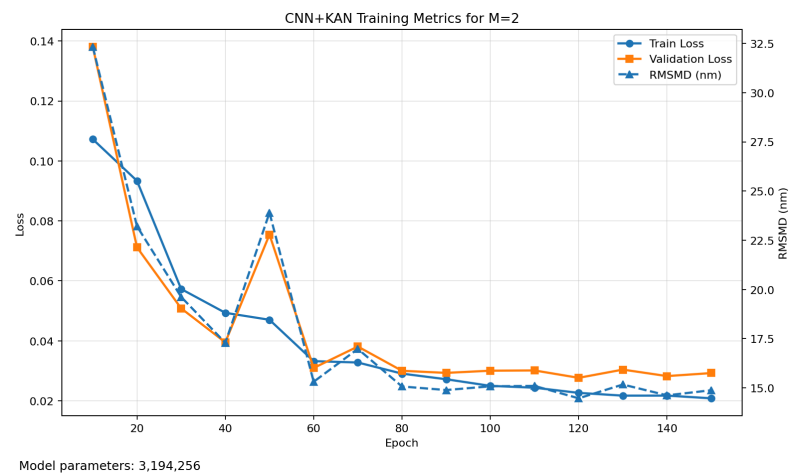
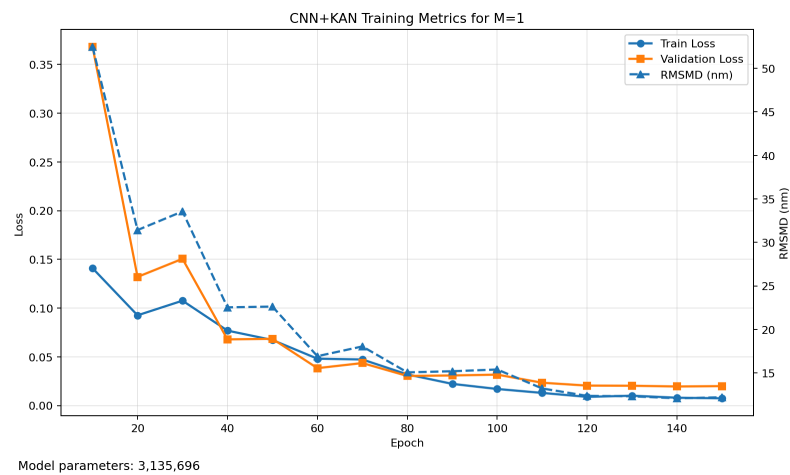


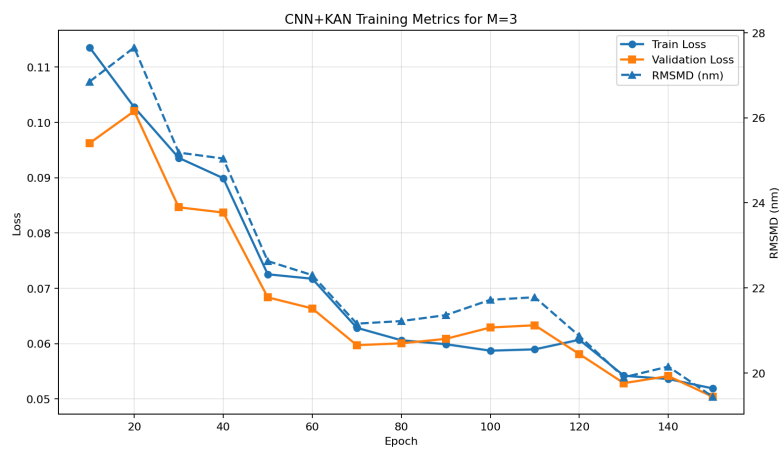
Training:



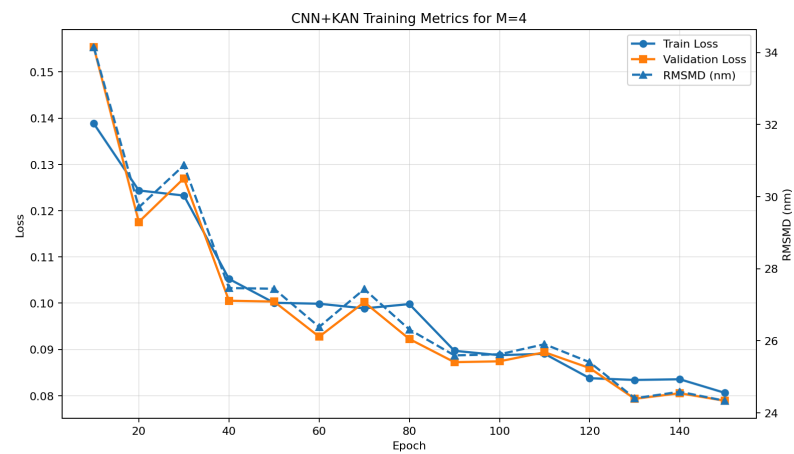


CNN+KAN Training:

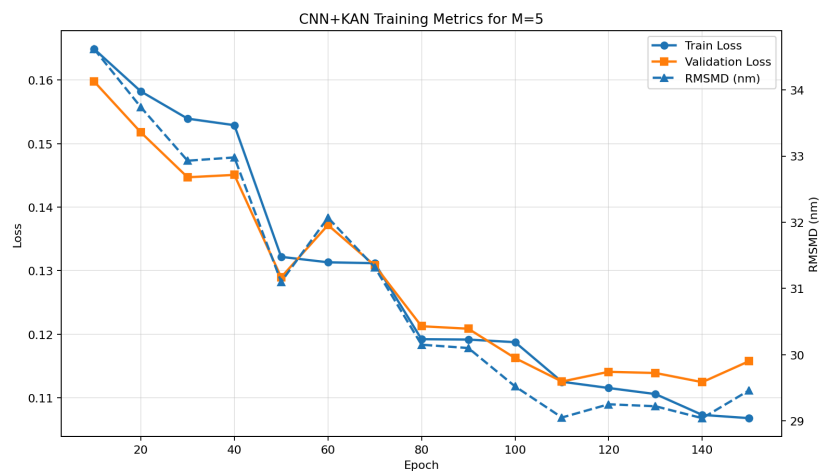




Model parameters: 4,813,296

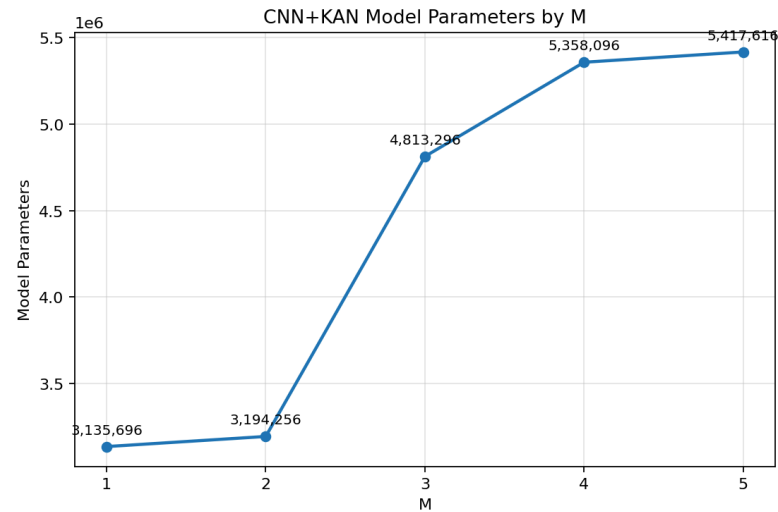
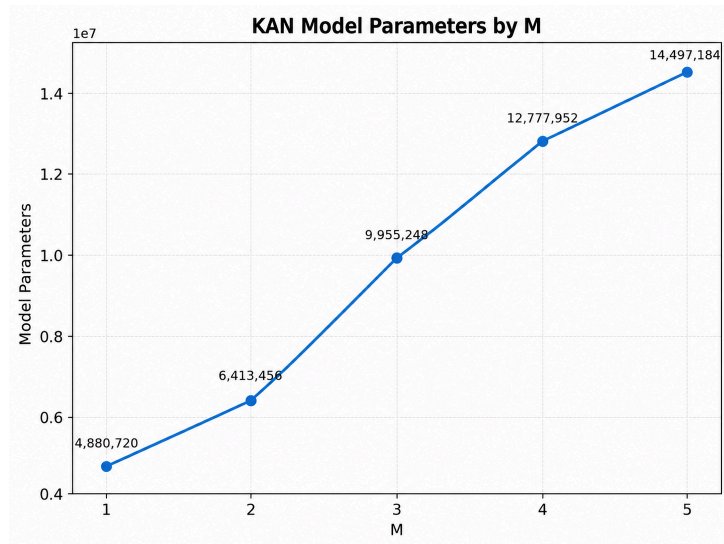


Model parameters: 5,358,096



Model parameters: 5,417,616

Number of Parameters:



Network: the first network architecture is the 21st network architecture I designed on the fixed emitter positions dataset:

The V21 localizer is a pure KAN in which all learnable parameters reside in B-spline activation functions on edges. Beyond standard unary edges, the architecture incorporates multi-arity interactions through pairwise tensor-product B-spline bases, enabling the network to capture second-order feature relationships while remaining within the KAN formulation. To prevent mode collapse in higher emitter counts caused by activations drifting outside the spline grid range and reducing the network to a linear model, LayerNorm is inserted between hidden layers. As LayerNorm performs no function learning and solely rescales activations into the active spline region, V21 is referred to as a KAN with inter-layer normalization.

The second network is the 6th network I designed for localizing emitters based on CNN+KAN. V6 localizer is a CNN+KAN hybrid that separates feature extraction from coordinate regression into two distinct stages. A lightweight CNN front-end processes the raw 7×7 fluorescence frame, provided as a three-channel input comprising pixel intensities and CoordConv-style normalized coordinate maps. Two convolutional blocks with batch normalization and ReLU activations progressively compress the spatial input from $3 \times 7 \times 7$ to a 288-dimensional feature vector, capturing the PSF spatial structure through weight sharing. This feature vector is then passed to a pure KAN head, which regresses the final sub-pixel emitter coordinates using learnable B-spline activations. By offloading spatial feature extraction to the CNN, the KAN head receives a compact, spatially-informed representation rather than raw pixels, allowing a shallower KAN to perform precise localization without the mode collapse observed in pure KAN architectures.

Result: but as you can see in the result visualization based on the network architecture KAN shows acceptable performance in comparison with CNN+KAN which is very promising.

